

Neural Clustering for Prefracted Mesh Generation in Real-time Object Destruction

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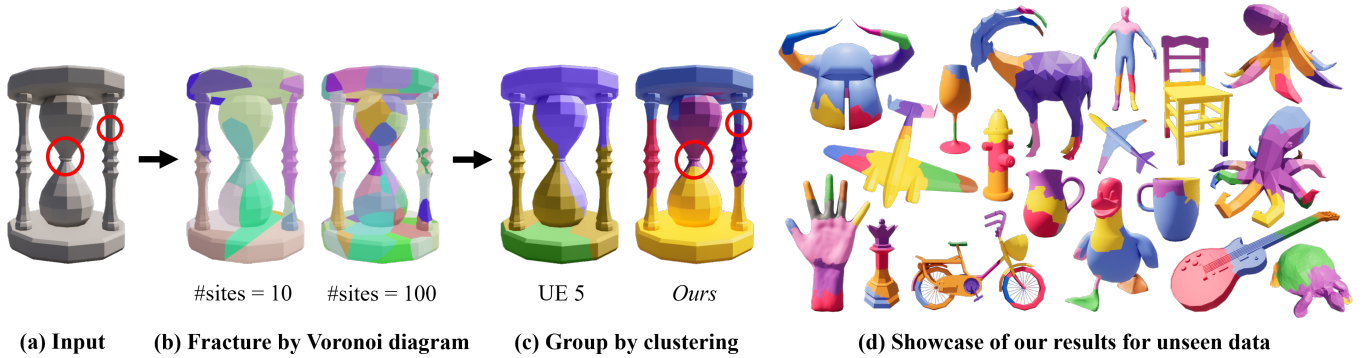


Figure 1: Prefracted mesh generation and results. Each piece and group is colored randomly. (a) Input mesh with highlighted structural weakness, thin points marked in red circles. (b) Prefracted meshes made solely from Voronoi diagrams, according to the number of sites. (c) Prefracted meshes refined from (b) with auto-clustering in Unreal Engine 5 (UE 5) and our method. Only ours reflects structural weaknesses at the designated positions, shown in red circles, whereas Voronoi-only and UE 5 do not. (d) Results of ours for unseen data from Objaverse [Deitke et al. 2023]. Each prefracted mesh considers the structure of object, for example assigning fractures to extremities such as wings, horns, and legs, which are prone to breakage.

Abstract

Prefracture method is a practical implementation for real-time object destruction that is hardly achievable within performance constraints, but can produce unrealistic results due to its heuristic nature. To mitigate it, we approach the clustering of prefracted mesh generation as an unordered segmentation on point cloud data, and propose leveraging the deep neural network trained on a physics-based dataset. Our novel paradigm successfully predicts the structural weakness of object that have been limited, exhibiting ready-to-use results with remarkable quality.

CCS Concepts

• **Computing methodologies** → **Shape inference**; *Physical simulation*; *Image segmentation*.

Keywords

Real-time object destruction, Prefracture, Point cloud segmentation

ACM Reference Format:

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1 Introduction

Real-time object destruction is challenging due to computational complexity. Fully physics-based simulations remain expensive despite recent efforts [Huang and Kanai 2023]. *Prefracture method* is a feasible implementation of real-time object destruction, which is adopted in practical applications like Unreal Engine 5 (UE 5). It eludes simulation at runtime by utilizing a pre-computed fractured version of the original mesh. When an impact is detected at runtime, the mesh is entirely or partially replaced with its prefracted counterpart. The remainder is rigidbody simulation with its fragments, incurring reasonable cost.

However, it relies on heuristics and thus can cause unrealistic destruction, as depicted in Figure 1. The generation of prefracted mesh typically involves breaking the mesh into fine pieces using Voronoi diagram, and then clustering them into larger pieces that are called *group* in this work. Various heuristics are used in this process for aesthetic results, but they often fail to reflect the structural weakness of object [Sellán et al. 2023]. Currently, this is manually corrected by an artist, which is very labor-intensive. This work addresses the issue using a deep neural network that learns from a physics-based object destruction dataset.

2 Method

We introduce the network into the clustering stage in prefractured mesh generation. Since it is difficult to handle the mesh directly, we gather the center of mass from pieces to construct point cloud. Our goal is clustering point cloud with ground truth group labels from datasets, which can be generalized to point cloud segmentation. We employ *Point Transformer* [Zhao et al. 2021] to predict a group label, as indicated in Figure 2. It provides U-net architecture with efficient self-attention on point cloud and shows competitive performance, leading to becoming a baseline for recent point cloud research. We use the default hyperparameter as proposed in the work, except for minor modifications to fit the shape of in/output.

However, the group label is distinct from the conventional segmentation label. We only focus on the identity between each point rather than the numerical value of group label. Namely, the group label is *unordered*. Hence the loss must be permutation-invariant in both the prediction and ground truth group label; i.e., remaining consistent if labels share the same group structure across points.

To resolve the unordered label, we design the loss based on pairwise identity, assuming point-to-point independence. For any points $x_i, x_j \in X$, we define the softmax of network output as the probability that each point belongs to a particular group $k \in G$, $p(g(x_i) = k)$, where $g(x_i)$ denotes the group label of x_i in an arbitrarily determined order. Assuming $p(g(x_i) = k)$ and $p(g(x_j) = k)$ are independent, $p(g(x_i) = g(x_j)) = \sum_k^G p(g(x_i) = k)p(g(x_j) = k)$ holds. From the ground truth group label in an arbitrarily determined order $y_i, y_j \in Y$, $p(y_i = y_j)$ is simply derived as the indicator function, which is 1 if equal, otherwise 0. We obtain a permutation-invariant loss by applying binary cross-entropy here:

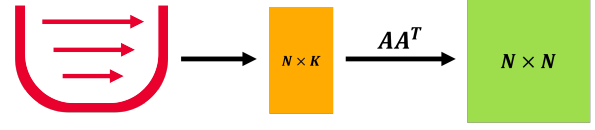
$$\begin{aligned} L(X, Y) = & \sum_i^N \sum_j^N [-p(y_i = y_j) \log \sum_k^G p(g(x_i) = k)p(g(x_j) = k) \\ & - (1 - p(y_i = y_j)) \log (1 - \sum_k^G p(g(x_i) = k)p(g(x_j) = k))] \\ & + \alpha \sum_k^G p(g(x_i) = k)p(g(x_j) = k) \end{aligned}$$

where $N = |X| = |Y|$. It can be efficiently computed by matrix multiplication as in Figure 2. We introduce the last term to regularize group clumping, and α is its weight, set to 0.1 in our experiment.

We sample $k \sim p(g(x_i))$ instead of $\arg\max_k p(g(x_i) = k)$ to get random outputs from the same input, which is often desired in prefracture method. $p(y_i = y_j)$ is the adjacency matrix of graph where vertices are X and edges are group identity relations. It consists of several clique components representing each group. Our network can be interpreted as a probabilistic model of this graph.

3 Experimental Results

As our method does not have any geometric prior itself, the quality of dataset is important, still there are few datasets on object destruction. We train the network with Breaking Bad [Sellán et al. 2022], which is a synthetic dataset generated by physics-based simulation [Sellán et al. 2023]. While it was originally proposed for 3D shape assembly, we flip its in/output pairs to supervise the network realistic object destruction. We break input mesh into fine pieces,



(a) Point Transformer (b) Group of each point (c) Pairwise group identity

Figure 2: Proposed network architecture. (a) Point Transformer [Zhao et al. 2021] as a backbone network. (b) The softmax of output as the probability that each point belongs to each group, where $N = |X|$ and $K = |G|$. (c) Matrix multiplication of (b) derives point-to-point group identity, providing permutation-invariant loss to group.

and label them depending on the output fragment mesh containing their center of mass. The number of fragments is concatenated to the input, allowing control over the number of groups at inference.

To enable practical application, we post-process the group labels to obtain the prefractured mesh. First, we merge the pieces that are in a group and share a face into a single mesh. Second, we split non-adjacent pieces within a group into separate groups. The resulting group mesh set is ready-to-use, e.g., in UE 5.

The evaluation was conducted on unseen data from Objaverse [Deitke et al. 2023]. Figure 1 presents comparisons (a, b, c) and the showcase (d). Our approach outperforms existing methods, particularly in detecting structural weaknesses.

4 Discussion

We propose a novel paradigm exploiting the deep neural network for real-time object destruction, identifying structural weaknesses. However, limitations include the scarcity of datasets and inadequacy of traditional segmentation benchmarks for evaluation. With emerging datasets like Fantastic Breaks [Lamb et al. 2023], which provides material and real-world data but lacks in scale, future research building on ours may achieve more realistic destruction.

Acknowledgments

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References

- Matt Deitke, Dustin Schwenk, Jordi Salvador, Luca Weihs, Oscar Michel, Eli VanderBilt, Ludwig Schmidt, Kiana Ehsani, Aniruddha Kembhavi, and Ali Farhadi. 2023. Objaverse: A universe of annotated 3d objects. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 13142–13153.
- Yuhang Huang and Takashi Kanai. 2023. DeepFracture: A Generative Approach for Predicting Brittle Fractures. *arXiv preprint arXiv:2310.13344* (2023).
- Nikolas Lamb, Cameron Palmer, Benjamin Molloy, Sean Banerjee, and Natasha Kholgade Banerjee. 2023. Fantastic breaks: A dataset of paired 3d scans of real-world broken objects and their complete counterparts. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 4681–4691.
- Silvia Sellán, Yun-Chun Chen, Ziyi Wu, Animesh Garg, and Alec Jacobson. 2022. Breaking bad: A dataset for geometric fracture and reassembly. *Advances in Neural Information Processing Systems* 35 (2022), 38885–38898.
- Silvia Sellán, Jack Luong, Leticia Mattos Da Silva, Aravind Ramakrishnan, Yuchuan Yang, and Alec Jacobson. 2023. Breaking good: Fracture modes for realtime destruction. *ACM Transactions on Graphics* 42, 1 (2023), 1–12.
- Hengshuang Zhao, Li Jiang, Jiaya Jia, Philip HS Torr, and Vladlen Koltun. 2021. Point transformer. In *Proceedings of the IEEE/CVF international conference on computer vision*. 16259–16268.